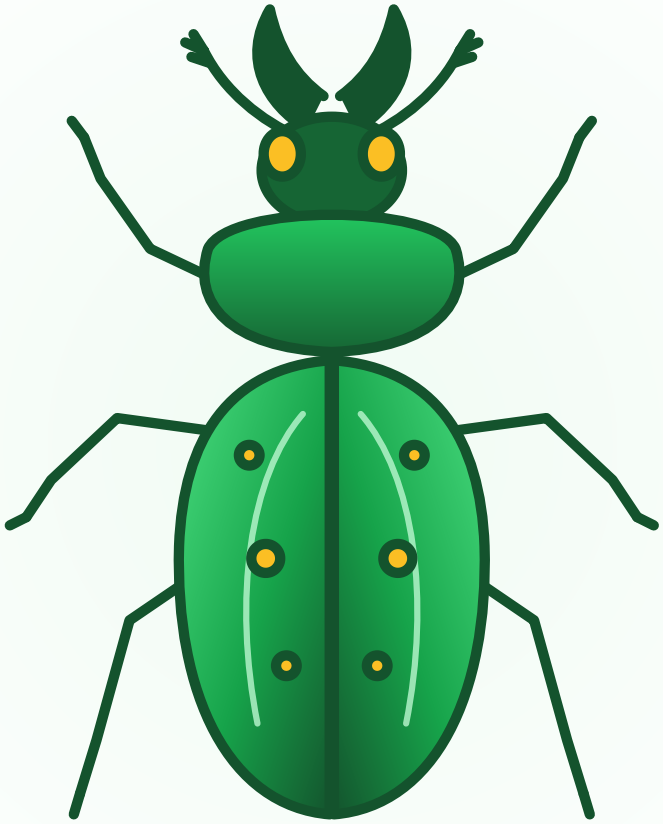
BUG DETECTIVE

Field Journal

Find it. Count it. Sketch it. Name it.



SPECIMEN LABEL

Field scientist: _____

Journal started: _____ **Bugs logged:** _____

If found, please return to the DDMES Science Olympiad binder shelf.

How a bug detective works

Scientists who study insects are called **entomologists** (en-toh-MOL-oh-jists). This journal is your field notebook. It goes in your SciOly binder, so bring it back every Friday.

1
SPOT
Look under leaves, on flowers, under rocks and logs, on windows and porch lights.

2
COUNT
Body parts, legs, antennae, pairs of wings. Use a hand lens.

3
SKETCH
Draw it big and label it. Add one hand-lens close-up.

4
KEY IT
Use the pocket key to find the class and the insect order.

5
LOG IT
Fill in an observation page and add it to your Life List.

The Field Scientist Code

- [ ] Look with your eyes and a hand lens first. Touch only if an adult says it is safe.
- [ ] **Never** pick up spiders, ticks, bees, wasps, fire ants, centipedes, or fuzzy caterpillars.
- [ ] Roll logs and rocks **toward** you, look, then roll them back gently.
- [ ] If you catch a bug in a cup, let it go where you found it within 10 minutes.
- [ ] A photo is a great specimen. Take pictures instead of collecting.
- [ ] Only explore near water with an adult. Wash your hands when you finish.

My field kit

- [ ] This journal and a pencil
- [ ] Colored pencils
- [ ] Hand lens (magnifying glass)
- [ ] Clear cup or bug viewer with a lid
- [ ] A phone or camera (with an adult)
- [ ] A watch or timer

Journal map

Tools	p. 3-7
Field observations	p. 8-15
Hunts and missions	p. 16-23
Life List and credits	p. 24-26

Need more observation pages? Ask a coach for copies of page 8, or print it again.

Bug hot spots

Porch lights at night
moths, beetles, and the spiders that hunt them

Under logs and rocks
millipedes, centipedes, pill bugs, beetle larvae

Flowers on a sunny day
bees, wasps, hoverflies, butterflies

Creeks and ponds
dragonflies, water striders, mayfly and caddisfly young

Milkweed and garden plants
aphids, ladybugs, caterpillars, stink bugs

Inside the house
silverfish, house centipedes, booklice (look, don't squish!)

Did you know? There are about 10 quintillion insects alive right now. That is about 1 billion insects for every person on Earth!

Is it an insect?

3 body parts

6 legs

2 antennae (1 pair)

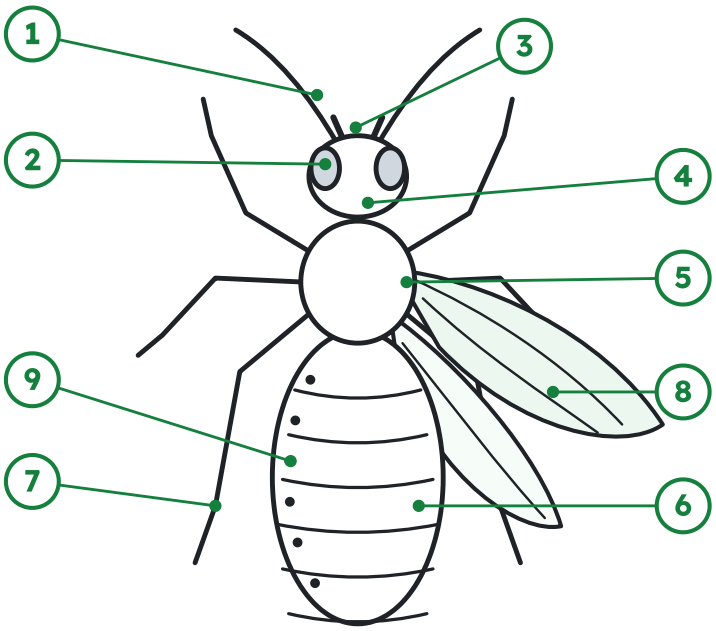
All 3? It's an **INSECT.**

All of these animals are **arthropods** (AR-throh-pods): they have a hard outside skeleton (exoskeleton), a body in segments, and legs that bend at joints. Insects are just 1 of the 5 classes.

Photo	Class	Body parts	Legs	Antennae
	**Insecta** *in-SEK-tuh* Ants, beetles, butterflies, flies	3	6	1 pair
	**Arachnida** *uh-RAK-nih-duh* Spiders, ticks, scorpions, daddy longlegs	2	8	none
	**Crustacea** *krus-TAY-shuh* Crabs, crayfish, shrimp, pill bugs	2 (most)	10 or more	2 pairs
	**Chilopoda** *ky-LOP-oh-duh* Centipedes (fast, flat, hunters)	many segments	1 pair per segment	1 pair
	**Diplopoda** *dih-PLOP-oh-duh* Millipedes (slow, round, eat dead leaves)	many segments	2 pairs per segment	1 pair (short)

Tricky ones: A pill bug is a **crustacean** (14 legs). A daddy longlegs is an **arachnid** (8 legs). A spider is **not** an insect!

Label the insect



1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

8 _____

9 _____

antenna **compound eye** **mouthparts** **head** **thorax** **abdomen** **leg** **wing** **spiracle**

Spiracles (SPEER-uh-kulls) are tiny breathing holes along the sides. Insects breathe through them, not their mouths!

Mouthpart match

Draw a line from each mouthpart to an insect that has it. Then act one out for a partner.

Chewing
jaws that bite and grind

Siphoning
a coiled tube for sipping nectar

Sponging
a spongy pad that mops up liquid

Piercing-sucking
a needle beak that pokes and sucks

- **mosquito**
- **house fly**
- **butterfly**
- **grasshopper**

The head senses and eats. The thorax moves (all 6 legs and the wings attach here). The abdomen digests and breathes.

The 17 insect orders

Lesson 2 orders (first two rows) and Lesson 3 orders (third row). *ptera* means "wing".



Coleoptera

koh-lee-OP-ter-uh · "sheath wing"

Beetles. Hard front wings meet in a straight line down the back.

Complete ☐ Spotted: _____



Lepidoptera

lep-ih-DOP-ter-uh · "scale wing"

Butterflies and moths. Scaly wings, coiled sipping tube.

Complete ☐ Spotted: _____



Hymenoptera

hy-men-OP-ter-uh · "membrane wing"

Bees, wasps, ants. Thin waist; back wings smaller.

Complete ☐ Spotted: _____



Diptera

DIP-ter-uh · "two wing"

Flies and mosquitoes. Only 1 pair of wings.

Complete ☐ Spotted: _____



Neuroptera

noo-ROP-ter-uh · "nerve wing"

Lacewings, antlions. Lacy veined wings held like a roof.

Complete ☐ Spotted: _____



Siphonaptera

sy-fon-AP-ter-uh · "tube, no wing"

Fleas. No wings, flat side to side, huge jumping legs.

Complete ☐ Spotted: _____



Orthoptera

or-THOP-ter-uh · "straight wing"

Grasshoppers, crickets. Big jumping back legs.

Incomplete ☐ Spotted: _____



Hemiptera

heh-MIP-ter-uh · "half wing"

True bugs, cicadas, aphids. Sucking beak; wings make an X.

Incomplete ☐ Spotted: _____



Blattodea

bla-TOH-dee-uh · "cockroach"

Roaches and termites. Flat, shield over the head.

Incomplete ☐ Spotted: _____

The 17 insect orders, continued

Lesson 3 orders (first row), then the Lesson 4 water insects and ancient insects.



Dermaptera

der-MAP-ter-uh · "skin wing"

Earwigs. Pincers (cerci) at the tail end.

Incomplete ☐ Spotted: _____



Phasmatodea

faz-muh-TOH-dee-uh · "ghost"

Stick and leaf insects. Look like twigs or leaves.

Incomplete ☐ Spotted: _____



Psocodea

so-KOH-dee-uh · "lice"

Lice and booklice. Tiny, flat, no wings.

Incomplete ☐ Spotted: _____



Odonata

oh-doh-NAH-tuh · "toothed"

Dragonflies, damselflies. Huge eyes, long thin body.

Incomplete ☐ Spotted: _____



Ephemeroptera

eh-fem-er-OP-ter-uh · "lives a day"

Mayflies. Wings straight up, 2 or 3 tail threads.

Incomplete ☐ Spotted: _____



Trichoptera

try-KOP-ter-uh · "hair wing"

Caddisflies. Hairy wings like a tent; larvae build cases.

Complete ☐ Spotted: _____



Archaeognatha

ar-kee-OG-nuh-thuh · "ancient jaw"

Jumping bristletails. Humped back, 3 tails, eyes touch.

None ☐ Spotted: _____



Zygentoma

zy-jen-TOH-muh · "joined insects"

Silverfish. Flat and silvery, 3 tails, eyes far apart.

None ☐ Spotted: _____

How to use

Check the box and write the date when you spot one of these orders **in real life**, at the museum, or in a new photo.

Complete egg, larva, pupa, adult

Incomplete egg, nymph, adult

None young look like tiny adults

Order Collector badge: spot 10 orders.

Pocket key: class and order

The same key as your Lesson 4 packet. Start at 1. Pick the choice that fits, then go to the step it names. Write the steps you used on your observation page (like 1, 2, 5, 9).

1	**a** 8 legs, no antennae: **ARACHNIDA**. **b** Not 8 legs: go to **2**.
2	**a** 6 legs and 3 body parts: **INSECTA**, go to **5**. **b** 10 or more legs: go to **3**.
3	**a** 2 pairs of antennae, hard shell, most live in water: **CRUSTACEA**. **b** 1 pair of antennae, long body of many segments: go to **4**.
4	**a** 1 pair of legs per segment, flat, fast: **CHILOPODA**. **b** 2 pairs of legs per segment, round, slow: **DIPLOPODA**.
5	**a** Has wings: go to **9**. **b** No wings: go to **6**.
6	**a** 3 long tail threads: go to **7**. **b** No tail threads: go to **8**.
7	**a** Humped back, big eyes that touch: **ARCHAEOGNATHA**. **b** Flat and silvery, small eyes far apart: **ZYGENTOMA**.
8	**a** Flat side to side, big jumping legs: **SIPHONAPTERA**. **b** Flat top to bottom, tiny, claws grip hair: **PSOCODEA**.
9	**a** Only 1 pair of wings: **DIPTERA**. **b** 2 pairs of wings: go to **10**.
10	**a** Front wings hard or leathery: go to **11**. **b** Front wings thin (clear, scaly, or hairy): go to **15**.
11	**a** Pincers at end of abdomen: **DERMAPTERA**. **b** No pincers: go to **12**.
12	**a** Hard front wings meet in a straight line down the back: **COLEOPTERA**. **b** Not a straight line: go to **13**.
13	**a** Sucking beak, wings make an X on the back: **HEMIPTERA**. **b** Chewing mouth: go to **14**.
14	**a** Big jumping back legs: **ORTHOPTERA**. **b** Looks like a stick or leaf: **PHASMATODEA**. **c** Flat and oval, shield over head: **BLATTODEA**.
15	**a** Wings covered in scales, coiled mouth tube: **LEPIDOPTERA**. **b** No scales: go to **16**.
16	**a** Hairy wings held like a tent: **TRICHOPTERA**. **b** Clear wings: go to **17**.
17	**a** Very long thin abdomen, huge eyes, tiny antennae: **ODONATA**. **b** Not: go to **18**.
18	**a** Wings held straight up, 2 or 3 long tail threads: **EPHEMEROPTERA**. **b** No tail threads: go to **19**.
19	**a** Thin waist, back wings smaller than front: **HYMENOPTERA**. **b** Lacy net of veins, wings held like a roof: **NEUROPTERA**.

Tricky ones: worker ants and termites have no wings (Hymenoptera, Blattodea). A hoverfly looks like a bee but has 1 pair of wings (Diptera). Cicadas and aphids have clear wings but a beak (Hemiptera).

Field Observation #1

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

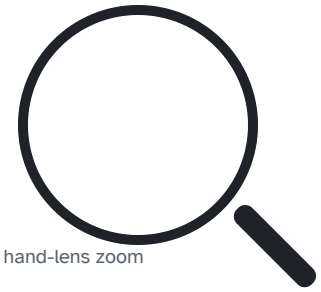
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

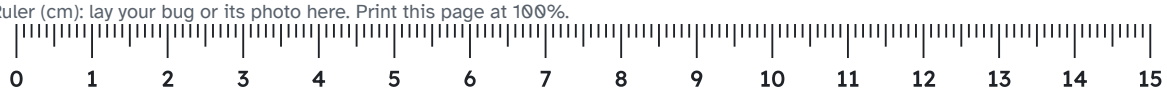
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #2

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

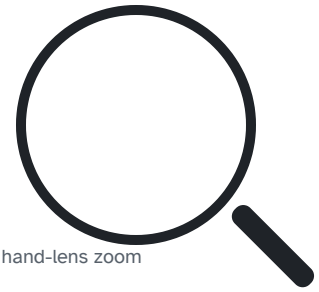
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

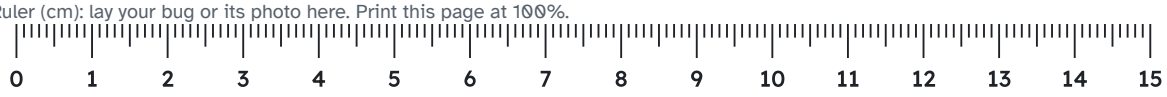
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #3

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

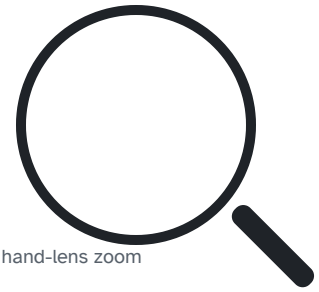
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

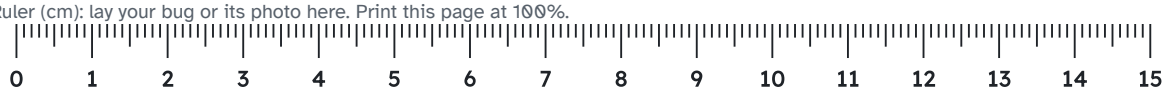
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #4

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

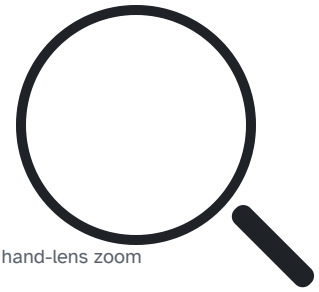
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

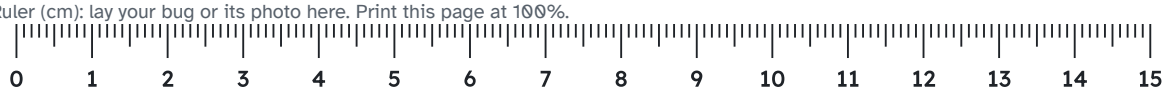
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #5

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

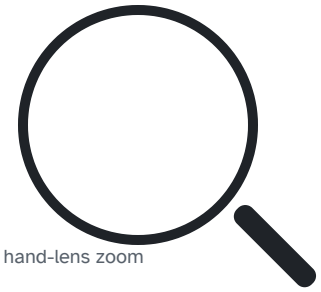
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

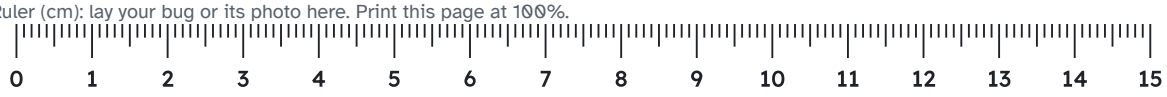
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #6

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

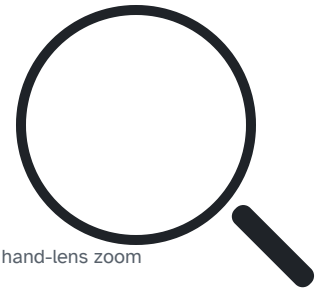
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

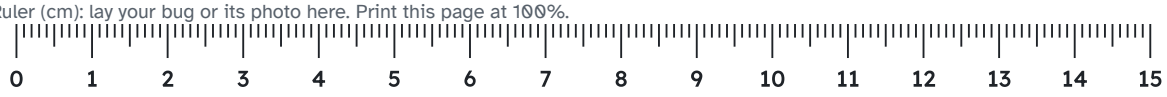
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #7

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

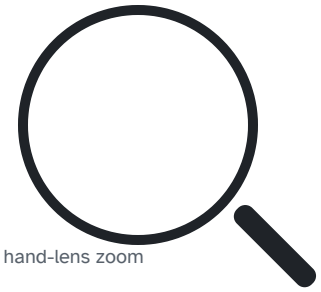
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

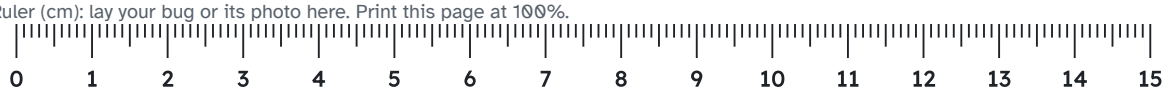
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Field Observation #8

How sure am I? ◯ Guess ◯ Pretty sure ◯ Certain

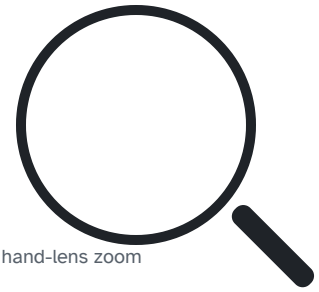
Date _____ **Time** _____ **Place** _____

☐ Sunny ☐ Partly cloudy ☐ Cloudy ☐ Rainy ☐ Windy **Temp: _____ °F**

Found:

☐ in soil or leaves ☐ on a plant or flower ☐ in or near water ☐ inside a building ☐ on an animal ☐ flying ☐ other

Sketch it BIG. Label 3 body parts.



hand-lens zoom

Real length: about _____ cm

COUNT IT

☐ body parts

☐ legs

☐ antennae

☐ pairs of wings

LIFE STAGE

☐ egg
☐ larva
☐ pupa
☐ nymph
☐ adult

MOUTHPARTS

☐ chewing
☐ siphoning
☐ sponging
☐ piercing-sucking
☐ can't tell

CLASS _____ **ORDER** _____ **COMMON NAME** _____

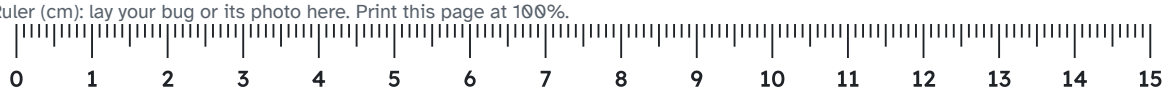
KEY STEPS I USED _____ **MY BEST CLUE** _____

What was it doing? ☐ eating ☐ flying ☐ crawling ☐ hiding ☐ hunting ☐ resting ☐ other

Helper, pest, or both? Why?

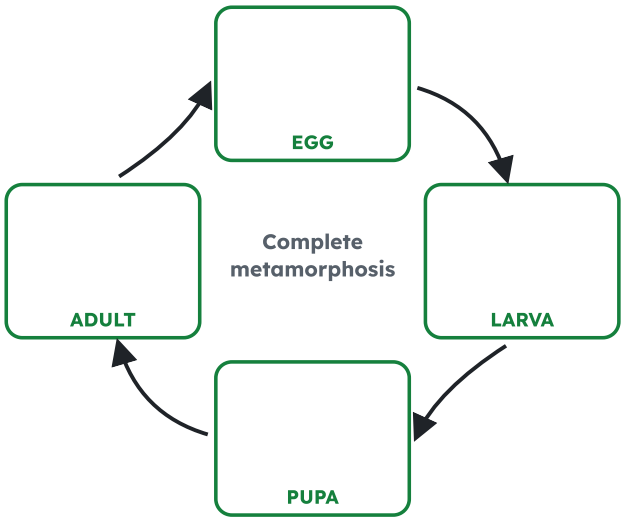
I wonder...

Ruler (cm): lay your bug or its photo here. Print this page at 100%.



Mealworm Metamorphosis Log

In Lesson 2 (Nov 6) the team gets a cup of mealworms. They are beetle larvae! Check the cup every time you come to practice and watch them turn into **darkling beetles** (order Coleoptera).



Draw each stage in its box. Where do mealworms fit?

Compare

Complete metamorphosis: egg, larva, pupa, adult. The baby looks nothing like the adult.

Incomplete: egg, nymph, adult. A nymph looks like a small adult without full wings (grasshoppers, dragonflies).

Molting: larvae shed their skin to grow. Look for empty skins in the cup!

Date	Larvae	Pupae	Beetles	What changed?

Why do you think the pupa does not move or eat?

Pollinator Patrol

Mission: pick one plant with flowers. Watch it for **10 minutes** without moving much. Make a tally mark each time an insect lands on a flower.

Date _____ **Time** _____ **Plant or flower color** _____

Visitor	Order or class	Tally marks	Total
Bees	Hymenoptera		
Wasps	Hymenoptera		
Flies and hoverflies	Diptera		
Butterflies and moths	Lepidoptera		
Beetles	Coleoptera		
Other or not sure			

> **Mimic alert!** A **hoverfly** wears bee stripes but has only 1 pair of wings. Count the wings before you tally.

Which visitor came most?

_____

Pollination means moving pollen from flower to flower so plants make fruit and seeds. How does the flower help the insect? How does the insect help the flower?

_____

_____

_____

Sketch the flower and one visitor

Log Roller Survey

Mission: with an adult, gently roll a log or big rock **toward you**. Count who lives underneath, then roll it back. Great in winter: many insects hide there as eggs, larvae, or adults until spring.

Date _____ **Place** _____ **Log or rock? Wet or dry?** _____

Visitor	Class	Tally marks	Total
Insects	Insecta: 6 legs		
Spiders, ticks, mites	Arachnida: 8 legs		
Centipedes	Chilopoda: 1 pair of legs per segment		
Millipedes	Diplopoda: 2 pairs per segment		
Pill bugs and sow bugs	Crustacea		
Worms, snails, slugs	NOT arthropods (no legs!)		

Decomposer food chain

Decomposers eat dead leaves and wood and turn them into soil. Draw a food chain with 3 living things you found (or know live here). Arrows point to the eater.



Which animal was the decomposer? How do you know?

Hide and seek: did anything use camouflage to hide? What did it look like?

Water Watch

Mission: some insects spend their young lives **under water**. Look in a creek or pond with an adult, at the museum, or in pictures. Mayfly nymphs and caddisfly larvae only live in clean water, so finding them is good news for the stream!

Young water insect	Order	Clean-water sign?	Found? How many?
Mayfly nymph gills along the sides, 2-3 tails	Ephemeroptera	Yes, very clean	
Caddisfly larva lives in a case of pebbles or sticks	Trichoptera	Yes, very clean	
Dragonfly nymph wide body, grabbing lip	Odonata	Okay water	
Damselfly nymph thin, 3 leaf-like tail gills	Odonata	Okay water	
Crayfish claws, 10 legs	Crustacea (not an insect)	Okay water	

Design a caddisfly case

Draw the case you built in Lesson 4 (gravel, lentils, or twigs). Why would a soft larva want armor?

Stream score

Did you find mayfly or caddisfly young?

☐ Yes: clean water! ☐ No ☐ Didn't look yet

Dragonfly or damselfly? A dragonfly rests with wings **open flat**. A damselfly rests with wings **closed over its back**.

What surprised me:

Museum Day: Insect Lab Tour

Saturday, November 7. Meet at 10 am at the Daily Planet Cafe. At 10:45 we tour the insect and arthropod research lab.

Before we go

Write 2 questions for the scientists.

1.

2.

Find one of each class

Class	What I saw
☐ Insecta	
☐ Arachnida	
☐ Crustacea	
☐ Chilopoda	
☐ Diplopoda	

Sketch the coolest specimen

Name: _____ Order: _____

The best thing a scientist told me

Could I be an entomologist?

Entomologists study bees for farms, track mosquitoes to stop disease, save butterflies, and solve crimes with flies. Which job would you want? Why?

Mystery Bug Challenge

Count first, then use your pocket key. Write the class for each one, and the order if it is an insect. Time yourself: _____ seconds. Then check the upside-down answers.

1



Class _____

Order _____

2



Class _____

Order _____

3



Class _____

Order _____

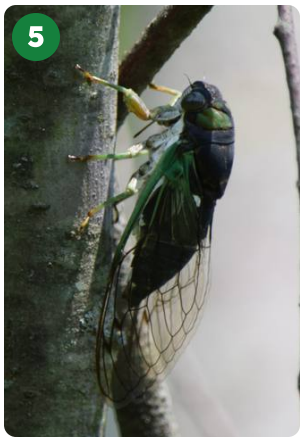
4



Class _____

Order _____

5



Class _____

Order _____

6



Class _____

Order _____

7



Class _____

Order _____

8



Class _____

Order _____

Answers: **1.** Scorpion: Arachnida (8 legs, no antennae) **2.** Luna moth: Insecta, Lepidoptera (scaly wings) **3.** Pill bug: Crustacea (lots of legs, 2 pairs of antennae) **4.** Paper wasp: Insecta, Hymenoptera (thin waist) **5.** Cicada: Insecta, Hemiptera (clear wings but a beak) **6.** Damselfly: Insecta, Odonata (wings closed over its back) **7.** Hercules beetle: Insecta, Coleoptera (straight line down the back) **8.** Field cricket: Insecta, Orthoptera (jumping back legs)

Field Missions

One mission for each Buzzworthy meeting. When you finish, show a coach or your Team Captain for initials, then color in the badge on the next page.

When	Mission	Badge	Done	Initials
After Fri Oct 2 Lesson 1	Arthropod hunt: log 3 arthropods from at least 2 different classes on observation pages.	Class Explorer	☐	
After Fri Nov 6 Lesson 2	Find an insect that does complete metamorphosis, in any life stage. Start the Mealworm Log.	Metamorphosis Master	☐	
Sat Nov 7 Museum	Fill in the Museum Day page and find all 5 classes.	Museum Explorer	☐	
After Fri Dec 11 Lesson 3	Do a Log Roller Survey. Find a hider using camouflage.	Log Roller	☐	
After Fri Jan 29 Lesson 4	Use the pocket key on 3 bugs and write the steps. Try Water Watch.	Key Master	☐	
Fri Feb 26 Lesson 5	Mystery Bug Challenge in under 2 minutes with 6 or more right.	Speed Spotter	☐	
Before Sat Mar 13 Tournament	Spot or study 10 of the 17 orders on the Order Collector pages.	Order Collector	☐	
Anytime Bonus	Pollinator Patrol: watch a flower for 10 minutes and tally the visitors.	Pollinator Patrol	☐	
Anytime Bonus	Water Watch: find (or look up) a mayfly or caddisfly young and fill in the page.	Water Watcher	☐	

Tournament day: Sat Mar 13, Chatham Regional

- ☐ 2 hand lenses
- ☐ Ref 1 (ID Binder) and Ref 2 (Ecology Binder)
- ☐ 2 sharp pencils

- **Count first:** legs, body parts, antennae, wings.
- Answer **both parts** of every station.
- Not sure? Write your best guess and move on when time is called.

My Bug Detective Badges

Color in each badge when you earn it. Earn all 9 to become a **Master Bug Detective**!



Class Explorer

Log 3 arthropods from 2 classes

Initials: _____



Metamorphosis Master

Find a complete-metamorphosis insect

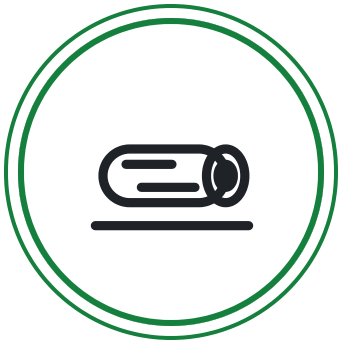
Initials: _____



Museum Explorer

Finish the Museum Day page

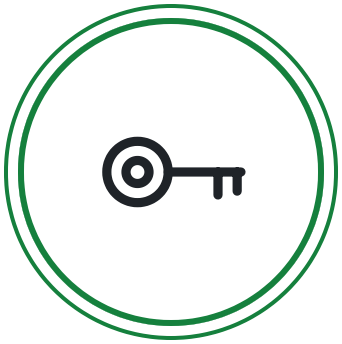
Initials: _____



Log Roller

Do a Log Roller Survey

Initials: _____



Key Master

Key out 3 bugs and show the steps

Initials: _____



Speed Spotter

6+ mystery bugs in under 2 minutes

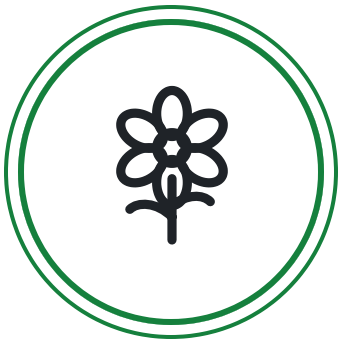
Initials: _____



Order Collector

Spot or study 10 of 17 orders

Initials: _____



Pollinator Patrol

Tally a flower for 10 minutes

Initials: _____



Water Watcher

Fill in the Water Watch page

Initials: _____

My Bug Life List

Every new kind of arthropod you identify goes here, even from the museum or a photo. Scientists call this a **life list**.

#	Date	Common name	Class	Order	Where
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					

My Bug Life List

Every new kind of arthropod you identify goes here, even from the museum or a photo. Scientists call this a **life list**.

#	Date	Common name	Class	Order	Where
25					
26					
27					
28					
29					
30					
31					
32					
33					
34					
35					
36					
37					
38					
39					
40					
41					
42					
43					
44					
45					
46					
47					
48					

Photo credits

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1. Insecta: Photo: ellen5, iNaturalist, CC-BY-NC
2. Arachnida: Photo: feistyone, iNaturalist, CC-BY
3. Crustacea: Photo: cullen, iNaturalist, CC-BY-NC
4. Chilopoda: Photo: macfoto, iNaturalist, CC-BY-NC
5. Diplopoda: Photo: allisonsuber, iNaturalist, CC-BY-NC
6. Coleoptera: Photo: dan_macneal, iNaturalist, CC-BY
7. Lepidoptera: Photo: musicwolf, iNaturalist, CC-BY-NC
8. Hymenoptera: Photo: molanic, iNaturalist, CC-BY-NC
9. Diptera: Photo: ibarzabal_j, iNaturalist, CC-BY-NC
10. Neuroptera: Photo: xpda, iNaturalist, CC-BY-SA
11. Siphonaptera: Photo: vuk, iNaturalist, CC-BY-NC
12. Orthoptera: Photo: glmory, iNaturalist, CC0
13. Hemiptera: Photo: tkoffel, iNaturalist, CC-BY-NC
14. Blattodea: Photo: curiousnomad, iNaturalist, CC-BY-NC
15. Dermaptera: Photo: gyrrlfalcon, iNaturalist, CC-BY-NC
16. Phasmatodea: Photo: jasondombroskie, iNaturalist, CC-BY-NC
17. Psocodea: Photo: datadan, iNaturalist, CC-BY
18. Odonata: Photo: jon_sullivan, iNaturalist, CC-BY
19. Ephemeroptera: Photo: thehaplesshiker, iNaturalist, CC-BY-NC
20. Trichoptera: Photo: cedricmondy, iNaturalist, CC-BY-NC
21. Archaeognatha: Photo: humanbyweight, iNaturalist, CC-BY-SA
22. Zygentoma: Photo: owenstrickland, iNaturalist, CC-BY-NC
23. Mystery bug 1: Photo: sheliahargis, iNaturalist, CC-BY-NC
24. Mystery bug 2: Photo: diohio1, iNaturalist, CC-BY-NC-SA
25. Mystery bug 3: Photo: berkshirenaturalist, iNaturalist, CC-BY-NC-ND
26. Mystery bug 4: Photo: sea-kangaroo, iNaturalist, CC-BY-NC-ND
27. Mystery bug 5: Photo: jamesjbond, iNaturalist, CC-BY-NC
28. Mystery bug 6: Photo: mchlfx, iNaturalist, CC-BY
29. Mystery bug 7: Photo: j_appleget, iNaturalist, CC-BY-NC
30. Mystery bug 8: Photo: thomasbarbin, iNaturalist, CC-BY-NC

Journal made for the DDMES Science Olympiad team, 2026-27. Order clues and the key match the Buzzworthy lesson packets and the 2027 Division A event rules (5 classes, 17 orders).